

- 30** The cathode of an electrolytic cell is a square piece of copper with dimensions 0.1 m x 0.1 m. The electrolyte is copper(II) sulfate.

Assume that each copper atom occupies a cube of length 3.0×10^{-12} m, the piece of copper has no thickness and that there is a uniform coverage.

How long will it take a current of 4.0 A to cover both sides of the piece of copper with new copper to a total of depth of 1000 atoms?

A 178 s

B 24.7 h

C 49.4 h

D 98.9 h